FINE GRAINED CAR DAMAGE DETECTION AND LOCALIZATION USING DEEP LEARNING

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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We hereby declare that the work entitled "**FINE GRAINED CAR DAMAGE DETECTION AND LOCALIZATION USING DEEP LEARNING**" is submitted in partial fulfillment of the requirement for the award of the degree in B.E, in University College of engineering, BIT Campus, Anna University, Tiruchirappalli. It is the record of my own work carried out during the academic year 2025 – 2026 under the supervision and guidance of **DR.S.USHA**, Assistant Professor(Sl.Grade), Department of Computer Science and Engineering, University College of Engineering, BIT Campus, Anna University, Tiruchirappalli. The extent and source of information are derived from the existing literature and have been indicated through the dissertation at the appropriate places. The matter embodied in this work is original and has not been submitted for the award of any other degree, either in this or any other university.

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ABSTRACT

Vehicle damage assessment is an important process in automobile repair, insurance claim verification, and service center inspection. Conventional vehicle inspection is performed manually by experts, and this approach is time-consuming, inconsistent, and prone to human error. Minor and fine-grained damages such as scratches, dents, damaged bumpers, broken mirrors, and cracked glass areas may be missed during manual observation. In addition, repair cost estimation often depends on individual judgement, which can lead to variation between inspectors and delay the claim settlement process.This project presents an automated web-based system for fine-grained car damage identification and localization using deep learning. The system accepts vehicle images from the user through image upload or camera capture. The images are processed using a deep-learning pipeline that first verifies whether the image contains a car, then classifies the vehicle as damaged or undamaged, and finally localizes the damaged vehicle parts using object detection. The implementation uses a Flask web application, MobileNetV2-based convolutional neural network classifiers for car detection and damage classification, and an Ultralytics YOLO object detection model for identifying affected parts such as hood, bumper, door, mirror, headlamp, windshield, tail light, trunk, grille, tire, and other important components.The proposed system also includes repair cost estimation using a predefined cost matrix. Based on the detected damaged parts and selected car brand, the system calculates an approximate repair cost and generates a detailed PDF assessment report. Owner details, car details, prediction results, damaged part list, estimated cost, warnings, and image-level results are stored using CSV-based storage. The generated report can also be sent to the customer through SMTP email configuration.The project provides a practical end-to-end solution for automated vehicle inspection. It reduces manual effort, improves consistency in damage assessment, supports faster insurance and repair workflows, and offers a user-friendly interface for vehicle owners, workshops, and insurance claim processors.

சுருக்கம்

வாகன சேத மதிப்பீடு என்பது வாகன பழுது பார்க்கும் செயல்முறை, காப்பீட்டு கோரிக்கை சரிபார்ப்பு மற்றும் சேவை மைய ஆய்வுகளில் முக்கியமான ஒன்றாகும். பாரம்பரியமாக, வாகன ஆய்வு நிபுணர்களால் கைமுறையாக மேற்கொள்ளப்படுகிறது. இந்த முறை நேரம் அதிகம் எடுத்துக்கொள்ளும், ஒரே மாதிரி துல்லியமற்றதாக இருக்கும் மற்றும் மனித பிழைகளுக்கு உட்பட்டதாகும். குறிப்பாக கீறல்கள், பள்ளங்கள், பம்பர் சேதம், கண்ணாடி உடைதல், மிரர் சேதம் போன்ற நுணுக்கமான சேதங்கள் கைமுறை ஆய்வில் கவனிக்கப்படாமல் போகும் வாய்ப்பு உள்ளது. மேலும், பழுது பார்க்கும் செலவு கணக்கீடு பெரும்பாலும் தனிநபர் மதிப்பீட்டின் அடிப்படையில் இருக்கும் காரணத்தால், ஆய்வாளர்களுக்கு இடையே வேறுபாடு ஏற்பட்டு, காப்பீட்டு செயல்முறை தாமதமாகும்.இந்த திட்டம் ஆழமான கற்றல் (Deep Learning) நுட்பங்களை பயன்படுத்தி நுணுக்கமான வாகன சேதங்களை தானியங்கி முறையில் கண்டறிந்து, அவற்றின் இடத்தை துல்லியமாக கண்டறியும் இணைய அடிப்படையிலான அமைப்பை முன்வைக்கிறது. இந்த அமைப்பு பயனரிடமிருந்து வாகன படங்களை பதிவேற்றம் அல்லது கேமரா மூலம் பெறுகிறது. அதன் பிறகு, படங்கள் பல கட்டங்களாக செயலாக்கப்படுகின்றன. முதலில், அந்த படம் வாகனத்தை கொண்டுள்ளதா என்பதை உறுதிப்படுத்துகிறது. பின்னர், வாகனம் சேதமடைந்ததா இல்லையா என்பதை வகைப்படுத்துகிறது. அதன் பிறகு, பொருள் கண்டறிதல் (Object Detection) நுட்பத்தின் மூலம் சேதமடைந்த பகுதிகளை கண்டறிகிறது.இந்த திட்டம் Flask அடிப்படையிலான இணைய பயன்பாடாக உருவாக்கப்பட்டுள்ளது. இதில் MobileNetV2 அடிப்படையிலான Convolutional Neural Network மாதிரிகள் வாகனத்தை கண்டறிதலும் சேத வகைப்பாடும் செய்ய பயன்படுத்தப்படுகின்றன. மேலும் Ultralytics YOLO மாதிரி பயன்படுத்தப்பட்டு ஹூட், பம்பர், கதவு, மிரர், ஹெட்லைட், விண்ட்ஷீல்ட், டெய்ல் லைட், டிரங்க், கிரில், டயர் போன்ற முக்கிய வாகன பகுதிகளில் ஏற்பட்ட சேதங்களை துல்லியமாக அடையாளம் காண்கிறது.மேலும், இந்த அமைப்பில் பழுது செலவு கணக்கீட்டு பகுதியும் இணைக்கப்பட்டுள்ளது. கண்டறியப்பட்ட சேத பகுதிகள் மற்றும் வாகனத்தின் பிராண்ட் அடிப்படையில் முன்கூட்டியே நிர்ணயிக்கப்பட்ட செலவுக் கட்டமைப்பைப் பயன்படுத்தி பழுது செலவு கணக்கிடப்படுகிறது. இந்த அமைப்பு உரிமையாளர் விவரங்கள், வாகன விவரங்கள், கண்டறிதல் முடிவுகள், சேதப்பட்ட பகுதிகள், மதிப்பிடப்பட்ட செலவு மற்றும் எச்சரிக்கை தகவல்களுடன் கூடிய விரிவான PDF அறிக்கையை உருவாக்குகிறது. இந்த தகவல்கள் CSV வடிவில் சேமிக்கப்படுகின்றன. மேலும், உருவாக்கப்பட்ட அறிக்கையை SMTP மின்னஞ்சல் அமைப்பின் மூலம் பயனருக்கு அனுப்பும் வசதியும் வழங்கப்பட்டுள்ளது.இந்த திட்டம் தானியங்கி வாகன ஆய்விற்கான முழுமையான தீர்வை வழங்குகிறது. இது மனித முயற்சியை குறைத்து, சேத மதிப்பீட்டின் துல்லியத்தை உயர்த்தி, காப்பீட்டு மற்றும் பழுது பார்க்கும் செயல்முறைகளை வேகமாக்குகிறது. மேலும், வாகன உரிமையாளர்கள், சேவை மையங்கள் மற்றும் காப்பீட்டு நிறுவனங்களுக்கு எளிதில் பயன்படுத்தக்கூடிய பயனர் நட்பு அமைப்பை வழங்குகிறது.

LIST OF TABLES

LIST OF FIGURES

LIST OF ABBREVIATIONS & ACRONYMS

AI - Artificial Intelligence

API - Application Programming Interface

CNN - Convolutional Neural Network

CSV - Comma Separated Values

DL - Deep Learning

GPU - Graphics Processing Unit

HTML - Hyper Text Markup Language

IoV - Internet of Vehicles

ML - Machine Learning

PDF - Portable Document Format

SMTP - Simple Mail Transfer Protocol

UI - User Interface

YOLO - You Only Look Once

CHAPTER 1
INTRODUCTION

Vehicle inspection is a necessary activity in automobile service centers, used-car evaluation, road accident analysis, and insurance claim processing. When a vehicle is damaged, the affected region must be identified accurately and the repair cost must be estimated in a fair and consistent manner. Traditionally, this task is performed by human inspectors who visually examine the vehicle, note damaged parts, estimate severity, and prepare reports for repair or insurance purposes.

Although manual inspection is widely used, it has several limitations. The process requires trained experts, consumes time, and may vary from one inspector to another. The judgement of damage severity and repair cost can be subjective. In busy service centers or large insurance claim departments, handling many inspection requests manually becomes difficult. Fine-grained damages such as scratches, small dents, broken lights, cracked glass, damaged side mirrors, and bumper deformation may be overlooked, especially when image quality or viewing angle is poor.

Recent developments in artificial intelligence and computer vision have created opportunities to automate this process. Deep learning models, especially Convolutional Neural Networks and object detection models, can learn visual patterns from large datasets and identify objects, parts, and damage regions from images. These techniques are already used in medical image analysis, surveillance, autonomous vehicles, and industrial inspection. Applying them to vehicle damage assessment can make the inspection process faster, more consistent, and more scalable.

The project "Fine Grained Car Damage Identification and Localization Using Deep Learning" focuses on developing a complete web-based vehicle damage assessment system. The system collects vehicle images from users, validates whether the image contains a car, detects whether the car is damaged, localizes the affected parts, estimates repair cost, stores assessment information, generates a PDF report, and supports email delivery of the report. The implemented solution combines deep learning with a user-friendly Flask interface to provide a practical automated inspection workflow.

1. Background

Automated vehicle damage assessment is important because it directly affects insurance claim settlement, vehicle repair planning, customer satisfaction, and operational efficiency. Vehicle owners expect quick and transparent estimates after an accident or damage event. Insurance companies require reliable evidence and consistent evaluation to reduce fraudulent claims and speed up approval.

1.1.1 Context of the Problem

Service centers need structured information about damaged components to plan spare parts, labour, and repair schedules.

The use of deep learning in this domain helps reduce dependence on purely manual inspection. A computer vision system can analyze uploaded images immediately and provide preliminary results within a short time. It can also maintain consistency because the same trained models and cost rules are applied across multiple users and claims. For organizations handling a large number of vehicle inspections, this approach improves scalability.

1.2 Problem Statement

Existing vehicle damage assessment methods depend heavily on manual inspection. This leads to delays, subjective judgement, inconsistent repair cost estimation, and difficulty in processing a large number of claims. Minor damages and part-level damages may be missed, and customers may not receive a clear explanation of the detected damage and estimated repair cost.

Therefore, there is a need for an automated system that can analyze car images, detect whether a car is present, classify the vehicle as damaged or undamaged, identify the affected vehicle parts, estimate repair cost using predefined rules, and generate a structured report. The system should be accessible through a web interface and should support practical inputs such as uploaded images and camera-captured images

1.3 Objectives

To design a user-friendly web interface for collecting owner details, car details, and vehicle images. To support both uploaded images and camera-captured images for vehicle assessment. To preprocess vehicle images and prepare them for deep learning model inference. To implement a car detection stage that verifies whether the uploaded image contains a vehicle. To implement a damage classification stage that classifies the vehicle as damaged or undamaged. To implement a part localization stage using YOLO object detection for detecting affected vehicle components. To estimate repair cost based on detected damaged parts and brand-specific cost values.

To store customer, vehicle, and assessment details using CSV-based storage. To

generate a PDF report containing assessment details, detected parts, and cost estimation.To support email delivery of the generated report using SMTP configuration

The scope of this project is limited to automated image-based car damage assessment. The system accepts car images and performs analysis using a multi-stage deep learning pipeline. It focuses on detecting the presence of a car, identifying whether damage exists, localizing damaged parts, and estimating repair cost from a predefined cost matrix.

The current implementation supports common vehicle parts including hood, front bumper, rear bumper, side mirror, headlamp, front door, rear door, fender, windshield, rear window, tire, tail light, grille, exhaust pipe, door handle, and trunk. The supported brands in the cost estimation module include Toyota, Honda, Hyundai, Suzuki, Tata, and Mahindra, with brand multipliers applied to base part costs.

The project is implemented as a Flask application and is suitable for local or institutional deployment. It is designed for preliminary assessment and decision support. Final repair decisions may still require expert validation, especially for severe structural damage, hidden mechanical damage, or poor-quality images.

1.3.3 Scope of the Project

The scope of this project is limited to automated image-based car damage assessment. The system accepts car images and performs analysis using a multi-stage deep learning pipeline. It focuses on detecting the presence of a car, identifying whether damage exists, localizing damaged parts, and estimating repair cost from a predefined cost matrix .The current implementation supports common vehicle parts including hood, front bumper, rear bumper, side mirror, headlamp, front door, rear door, fender, windshield, rear window, tire, tail light, grille, exhaust pipe, door handle, and trunk. The supported brands in the cost estimation module include Toyota, Honda, Hyundai, Suzuki, Tata, and Mahindra, with brand multipliers applied to base part costs.

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Chapter 2

Literature Review

2.1 Introduction

The literature survey studies existing approaches related to vehicle detection, vehicle damage classification, object detection, segmentation, accident detection, and automated inspection. Early systems used traditional image processing and machine learning methods, while recent systems mainly use deep learning. CNN-based classification models are effective for identifying whether damage exists, while object detection and segmentation models are useful for localizing damaged regions and identifying affected parts.This chapter reviews selected works relevant to the project and identifies limitations that motivated the proposed system.

2.2 Review of Existing Systems and/ Papers

The existing system mainly depends on manual vehicle inspection by insurance agents or service center staff, which takes more time and effort. Its accuracy depends on human experience, so damage identification, cost estimation, and report preparation may vary or be delayed.

2.2.1 TITLE: DeepCrash: Deep Learning-Based Accident Detection

AUTHOR: Wan-Jung Chang, Liang-Bi Chen, and Ke-Yu Su proposed DeepCrash, a deep learning-based Internet of Vehicles system for accident detection and emergency notification.

The system used vehicle sensor information and deep learning to detect accident events and trigger alerts. The work is useful for emergency response, but it focuses mainly on collision detection rather than detailed post-accident vehicle damage assessment. It does not identify damaged vehicle parts or estimate repair cost.

2.2.2 TITLE: Vehicle Damage Detection Using Improved Mask R-CNN

AUTHOR: Qinghui Zhang, Xianing Chang, and Shanfen Bian proposed an improved Mask R-CNN based method for vehicle damage detection and segmentation.

Mask R-CNN is capable of instance segmentation, which helps identify damaged regions more precisely. However, the system depends on the availability of a sufficiently large and diverse dataset. Very small scratches or subtle damage regions may still be difficult to segment accurately. The approach also focuses more on region segmentation than on complete cost estimation and report generation.

2.2.3 TITLE: Car Damage Detection and Classification

AUTHOR: Phyu Mar Kyu and Kuntpong Woraratpanya presented a car damage

detection and classification approach using CNN architectures such as VGG16 and VGG19 with transfer learning.

Their method demonstrated that deep learning can classify vehicle damage from images effectively. However, the system faced limitations such as overfitting, limited dataset availability, and difficulty in distinguishing fine-grained damage severity levels. It also did not provide a full web-based assessment workflow with part-level cost estimation

2.2.4 TITLE: Automated Vehicle Inspection Using Deep Learning

AUTHOR: Mohamed Mostafa Fouad and co-authors proposed an automated vehicle inspection model using deep learning.

The system used DenseNet-based models for classifying vehicle damage. DenseNet architectures can achieve strong feature extraction performance, but they may require more memory and computing resources. The work mainly focused on classification and did not fully address detailed damage localization, user workflow, PDF report generation, and email-based delivery.

2.2.5 TITLE: Multi-View Damage Inspection Using Single-View Damage Projection

AUTHOR: R. E. van Ruitenbeek and S. Bhulai studied multi-view damage inspection using single-view damage projection.

Their approach used object detection and 3D vehicle modeling to project visible damage information across vehicle views. This is useful for understanding spatial relationships between vehicle parts. However, the accuracy depends on the quality of the 3D model and the consistency between model geometry and real vehicle shapes. This makes the system complex for general web-based deployment.

2.2.6 TITLE: Vehicle Detection Systems for Intelligent Driving

AUTHOR: Rahib Abiyev and Murat Arslan discussed vehicle detection systems for intelligent driving using deep convolutional neural networks.

Vehicle detection is an important foundation for intelligent transportation systems and autonomous driving. However, this work focuses on detecting vehicles in road environments rather than identifying damage or estimating repair cost. It supports the relevance of CNN-based vehicle recognition but does not solve the complete damage assessment problem.

2.3 Comparative Analysis

Table 2.1 Comparative Analysis

S. No.	Title	Method / Technology	Limitations	Relevance to Proposed Work
1	DeepCrash	IoV, CNN, DenseNet	Focuses on accident detection, not part-level damage	Shows use of deep learning in vehicle safety
2	Improved Mask R-CNN for Vehicle Damage	Mask R-CNN, ResNet, FPN	Requires larger dataset; subtle damages may be missed	Supports localization-based damage analysis
3	Car Damage Detection and Classification	VGG16, VGG19, CNN	Overfitting; limited fine-grained severity classification	Supports CNN-based damage classification
4	Automated Vehicle Inspection	DenseNet-169	High memory usage; limited localization	Shows deep learning usefulness for inspection
5	Multi-View Damage Inspection	YOLOv5, 3D projection	Depends on accurate 3D vehicle models	Supports object detection for damage inspection
6	Vehicle Detection for Intelligent Driving	CNN, sliding window	Detects vehicles, not damage	Supports car detection as first-stage validation

Chapter 3
Proposed System

3.1 Existing System

In the existing manual system, vehicle damage assessment is performed by trained inspectors or service center staff. The inspector observes the vehicle physically or through images and records visible damages.

3.1.1 Working of Existing Systems

Based on experience, the inspector identifies affected parts and estimates repair cost. In insurance workflows, this information may be verified again by claim officers before approval.

Some existing automated systems use traditional image processing or machine learning to classify images. These systems usually require manual feature extraction and may not perform well under different lighting conditions, backgrounds, camera angles, or vehicle models. Other systems use deep learning but focus only on binary classification such as damaged or undamaged. Such systems do not provide part-level information or cost estimation.

3.1.2 Drawbacks of Existing Systems

The major drawbacks of the existing system are:

Manual inspection takes more time and delays insurance claim processing.Human judgement can vary between inspectors.Minor and fine-grained damages may be missed.Many systems do not localize damaged vehicle parts.Repair cost estimation is not standardized.

Existing classification-only models do not provide complete repair decision support.Many solutions lack a user-friendly web interface.Report generation and email delivery are often not integrated.Manual record maintenance can cause data loss or inconsistency.Handling many claims at the same time is difficult without automation.

3.2 Proposed System

The proposed system is an intelligent web-based car damage assessment application. It uses deep learning and computer vision techniques to identify car damage from images and estimate repair cost.

3.2.1 Overview of the Proposed System

The system follows a multi-stage pipeline:

Input Image -> Car Detection -> Damage Classification -> Damaged Part Localization -> Cost Estimation -> PDF Report Generation -> Email Delivery

The first stage checks whether the uploaded image contains a car. This prevents invalid images from being processed further. The second stage classifies the detected car image as damaged or undamaged. If damage is identified, the third stage applies YOLO object detection to localize damaged parts. The detected parts are passed to a cost estimation module that calculates approximate repair cost using a brand-specific cost matrix.The web application guides the user through a step-by-step process. The user first enters owner details, then car details, then uploads or captures vehicle images. After processing, the system displays the result page with damage status, damaged parts, estimated cost, warnings, image-level results, and report download option. The report can be sent to the owner through email.

3.2.2 System Concept

The proposed system is an automated car damage detection and repair cost estimation system. It uses machine learning and image processing techniques to analyze vehicle images and identify whether the car is damaged or undamaged. If damage is detected, the system further identifies the affected car parts such as bumper, door, bonnet, fender, light, or windshield.

The system reduces the need for manual vehicle inspection by allowing users to upload a car image and receive damage-related information automatically. Based on the detected damaged parts, the system estimates the approximate repair cost using a predefined cost matrix. This makes the process faster, more consistent, and useful for insurance claim processing, service centers, and vehicle owners.

3.2.3 Key Features

The key features of the proposed system include automatic car image upload and analysis, where the user can provide a vehicle image for inspection. The system first detects whether the uploaded image contains a car and then classifies it as damaged or undamaged. If damage is found, it identifies the affected car parts such as bumper, door, bonnet, fender, light, and windshield. Based on the detected damaged parts, the system estimates the approximate repair cost using a predefined cost matrix. It reduces manual inspection effort, provides faster and more consistent results, and offers a user-friendly interface for viewing the final output. This system is useful for vehicle owners, service centers, and insurance companies because it supports quick damage assessment and repair cost estimation.

3.2.4 Advantages

The main advantage of this project is that it automates the process of car damage detection and cost estimation using deep learning.

It reduces the need for manual inspection by identifying whether a vehicle is damaged or undamaged and detecting the specific damaged parts such as bumper, door, bonnet, fender, light, and windshield.

The system saves time, improves consistency, and reduces human error in the assessment process. It also provides an estimated repair cost based on the detected damage, which is useful for vehicle owners, service centers, and insurance companies. Since the project includes a user-friendly web interface, image upload or camera capture, PDF report generation, and email delivery, it makes the complete damage assessment process faster, easier, and more convenient

3.3 System Architecture

3.3.1 Block Diagram

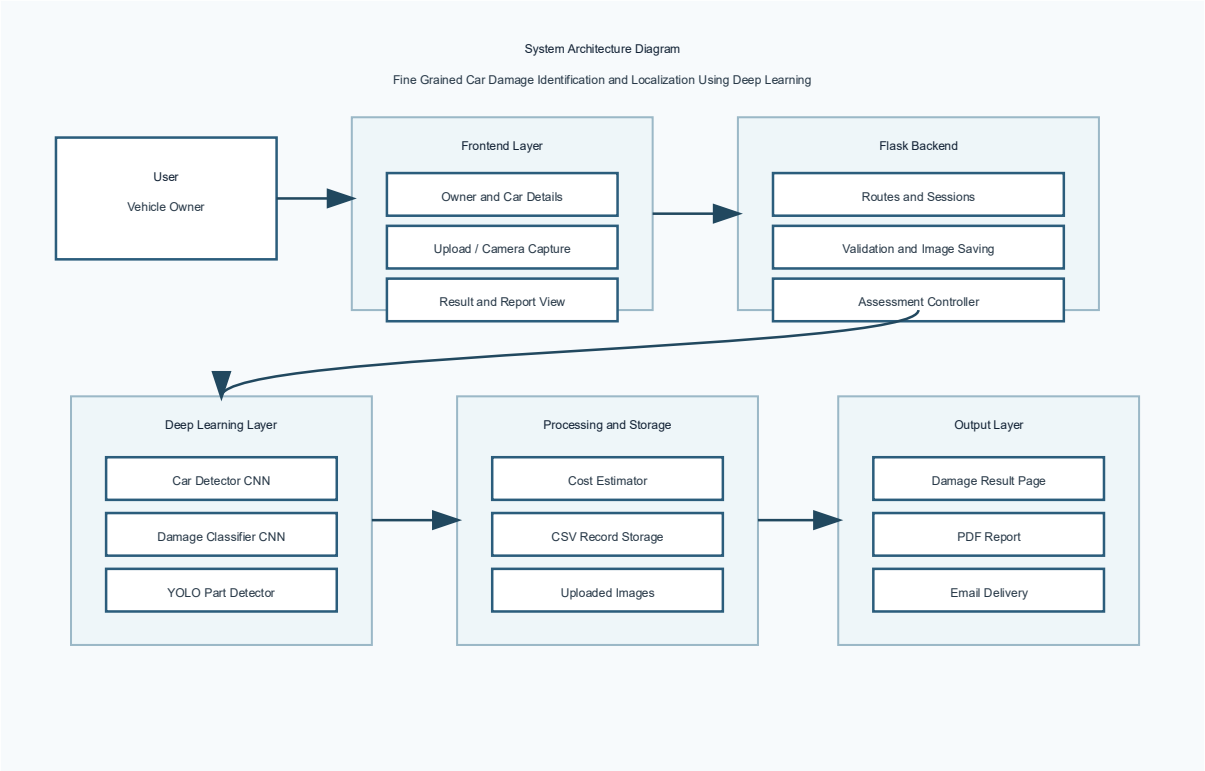

Figure (1) Block Diagram

3.3.2 Explanation of Modules

The architecture of the proposed system contains the following layers:

1. User Interface Layer

The user interface is implemented using HTML, CSS, JavaScript, and Flask templates. It provides pages for home, owner details, car details, assessment upload, camera capture, and result display.

2. Application Layer

The Flask backend handles routing, session management, form validation, image upload handling, camera image saving, assessment workflow, report generation, CSV storage, and email sending.

3. Deep Learning Layer

The deep learning layer contains the model manager. It loads trained model weights from the models/trained directory. The pipeline uses MobileNetV2-based classifiers for car detection and damage classification. YOLO is used for part detection and localization.

4. Cost Estimation Layer

The cost estimation module reads a cost matrix from dataset/cost_matrix.csv. It calculates part-wise and total repair cost based on the selected car brand and detected damaged parts.

5. Storage and Report Layer

CSV storage is used to save assessment records in data.csv. ReportLab is used to generate PDF reports. SMTP configuration is used to send generated reports through email.

3.4 Methodology

The methodology followed by the proposed system is described below.

3.4.1 Image Acquisition

Vehicle images are acquired from the user through two methods. The user may upload existing image files or capture images using the browser camera. This makes the system flexible for both desktop and mobile users.

3.4.2 Input Validation

The system validates owner details, car details, and assessment assets. Owner details include name, address, phone number, and email. Car details include car number, brand, model, and year. The assessment page verifies that at least one image is provided before processing.

3.4.3 Image Storage

For every assessment, a unique assessment ID is generated. Uploaded and captured images are stored under a separate folder inside the static uploads directory. This keeps each assessment organized and helps link input images with the generated report.

3.4.4 Car Detection

The first model checks whether the input image contains a car. This stage reduces false processing of invalid images. If no car is detected, the system returns a clear message and avoids unnecessary damage estimation.

3.4.5 Damage Classification

If a car is detected, the damage classifier predicts whether the vehicle is damaged or undamaged. The classifier uses a CNN-based architecture, with MobileNetV2 as the default backbone. The model outputs a damage label and confidence value.

3.4.6 Damaged Part Localization

If the vehicle is classified as damaged, the YOLO object detection model is used to identify affected vehicle parts. YOLO returns detected part labels, confidence scores, and bounding box coordinates. The system supports multiple vehicle components such as bumper, hood, doors, mirrors, lights, windows, tires, grille, exhaust pipe, door handle, and trunk.

3.4.7 Cost Estimation

The detected damaged parts are sent to the cost estimation module. The system reads part costs from the cost matrix and applies brand-based cost values. The total estimated cost is calculated by summing the cost of all detected damaged parts.

3.4.8 Result Generation

The result page displays owner details, car details, damage status, damaged parts, estimated cost, warnings, and image-level prediction results. If no damage is found, the system reports that the car is undamaged and no repair is required.

3.4.9 PDF Report Generation

A PDF report is generated using ReportLab. The report contains the assessment summary, owner information, car information, damage status, damaged parts, estimated repair cost, and uploaded image references.

3.4.10 Email Delivery

The generated PDF report can be sent to the owner through email using SMTP settings. SMTP details can be provided through environment variables or form input.

3.4.11 Flowchart of the Proposed System

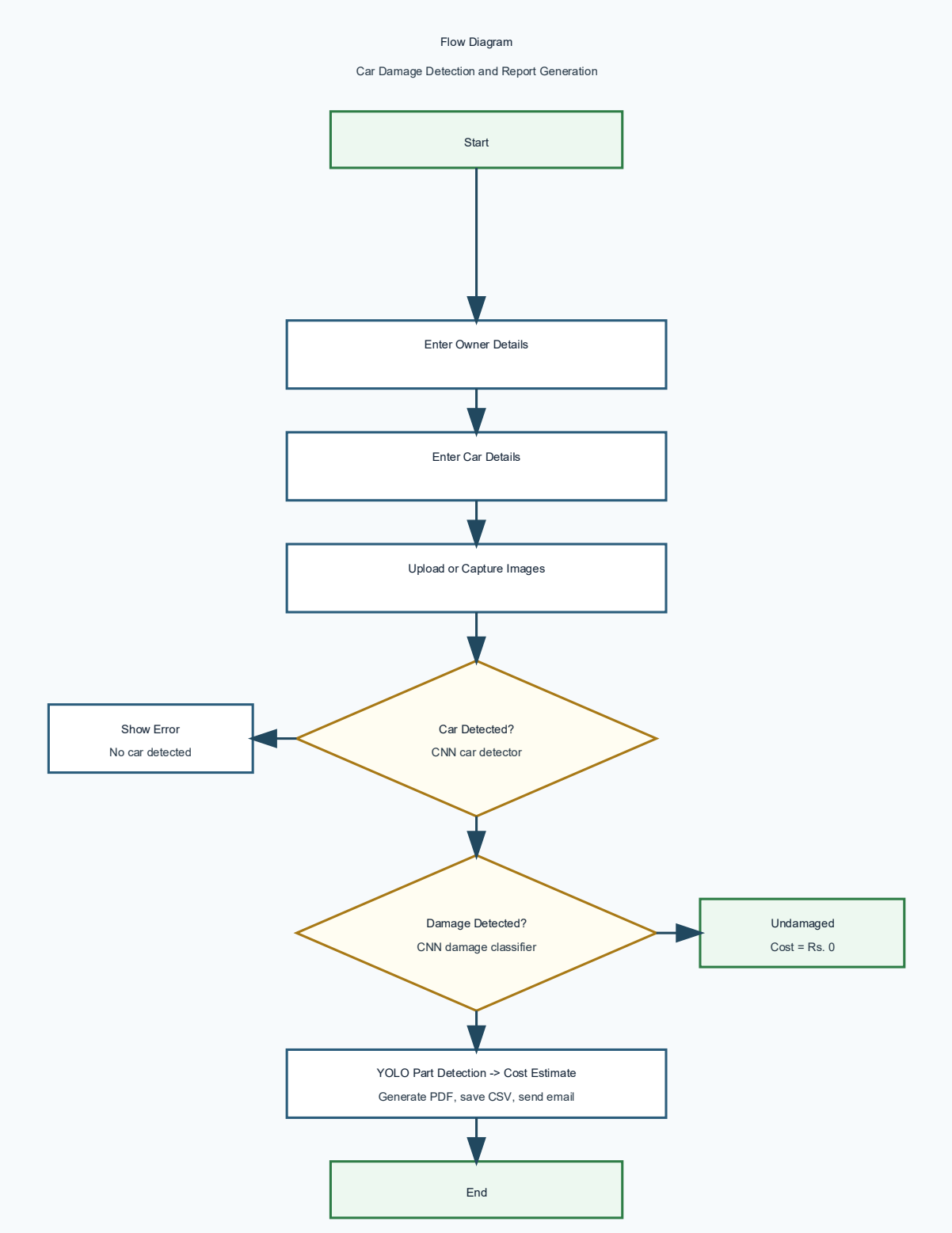

Figure (2) Flowchart

3.5 Module Description

The proposed system is designed using a modular architecture, where each module performs a specific task in the overall damage detection process. This modular design improves system clarity, maintainability, and scalability.

3.5.1 Owner Details Module

This module collects customer information such as name, address, phone number, and email. It validates the input and stores the details in the session for the current assessment.

3.5.2 Car Details Module

This module collects car number, brand, model, and manufacturing year. The brand selection is connected to the cost estimator so that repair cost can be calculated according to available brand-specific values.

3.5.3 Assessment Module

This module accepts uploaded images and camera-captured images. It creates an assessment folder, saves the images, and sends the file paths to the model manager for prediction.

3.5.4 Model Manager Module

The model manager controls the deep learning inference pipeline. It checks dependencies, loads trained weights, prepares image batches, predicts car presence, predicts damage status, and runs YOLO part detection.

3.5.5 Cost Estimator Module

The cost estimator reads the cost matrix and provides part-wise and total repair cost. It uses default base costs and brand multipliers when required.

3.5.6 CSV Storage Module

The CSV storage module saves assessment records such as owner name, contact details, car information, damage status, damaged parts, and estimated cost. This creates a simple record history without requiring a full database server.

3.5.7 Report Generator Module

The report generator creates a PDF report for each assessment. The report can be downloaded from the result page and attached to an email.

3.5.8 Email Sender Module

The email module sends the generated PDF report to the owner or another recipient using SMTP. This improves usability for customers and claim processing.

Chapter 4
Implementation & Results

4.1 Tools & Technologies

The project uses Python and Flask for backend development, HTML, CSS, and JavaScript for frontend design, and deep learning technologies such as TensorFlow/Keras and YOLO for vehicle damage detection. Supporting libraries like OpenCV, Pandas, NumPy, and Matplotlib help in image processing, data handling, and model performance visualization. Together, these tools make the system suitable for automated vehicle damage detection, cost estimation, and report generation.

4.1.1 Software

Table 4.1 Software Requirements

S. No.	Software	Purpose
1	Windows Operating System	Development and execution platform
2	Python	Backend programming and model integration
3	Flask	Web application framework
4	TensorFlow	CNN model training and inference
5	Ultralytics YOLO	Damaged part localization
6	OpenCV	Image reading, resizing, and preprocessing
7	NumPy	Numerical processing
8	ReportLab	PDF report generation
9	VS Code	Development environment
10	CSV	Record storage and cost matrix

4.1.2 Hardware

Table 4.2 Hardware Requirements

S. No.	Hardware	Minimum Requirement
1	Processor	Intel i5 or above
2	RAM	Minimum 8 GB
3	Storage	256 GB SSD or above
4	Camera	Smartphone camera
5	GPU	Optional, recommended for faster model training
6	Internet	Required for email delivery and dependency setup

4.2 System Implementation

4.2.1 Home Page Implementation

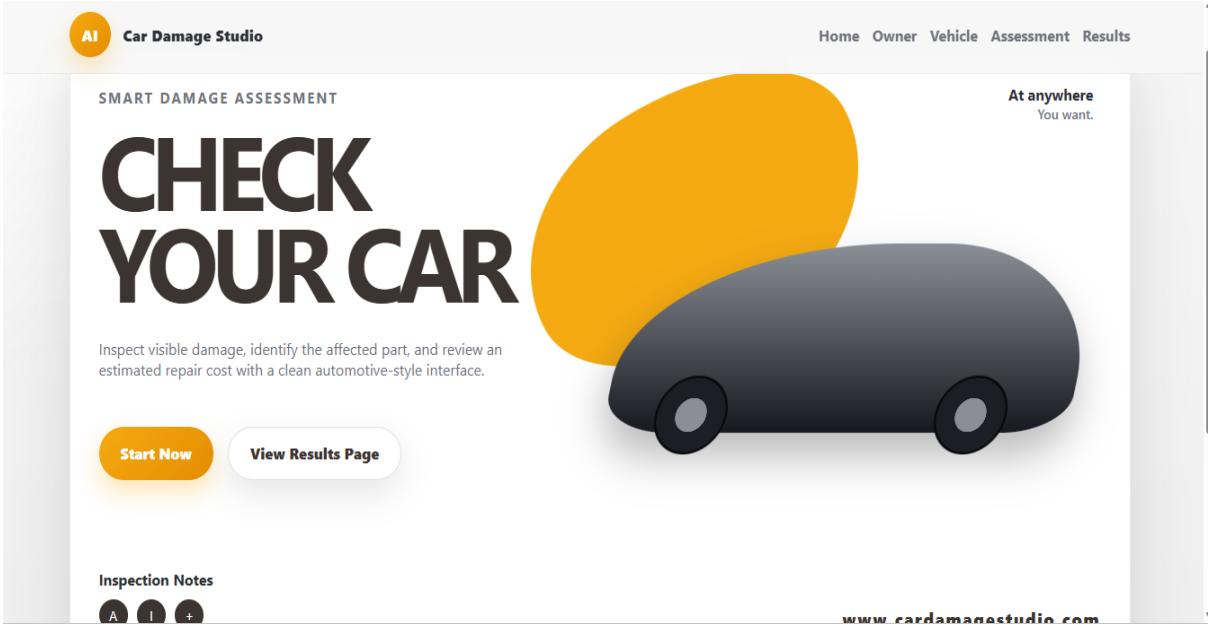

Figure (3) HomePage

The Home page acts as the starting point of the Car Damage Studio system. It introduces the purpose of the application and allows the user to begin the damage assessment process.

The page contains a clear title, Check Your Car, which represents the main function of the system. It also includes a short description explaining that the system can inspect visible damage, identify the affected vehicle part, and estimate the repair cost. The Start Now button redirects the user to the first workflow step, where owner details are collected. The View Results Page button allows the user to directly view previously generated assessment results.

The design uses a clean automotive-style interface with a car illustration, navigation bar, and action buttons. This makes the system simple, attractive, and easy to understand for users.

4.2.2 Owner Details Implementation

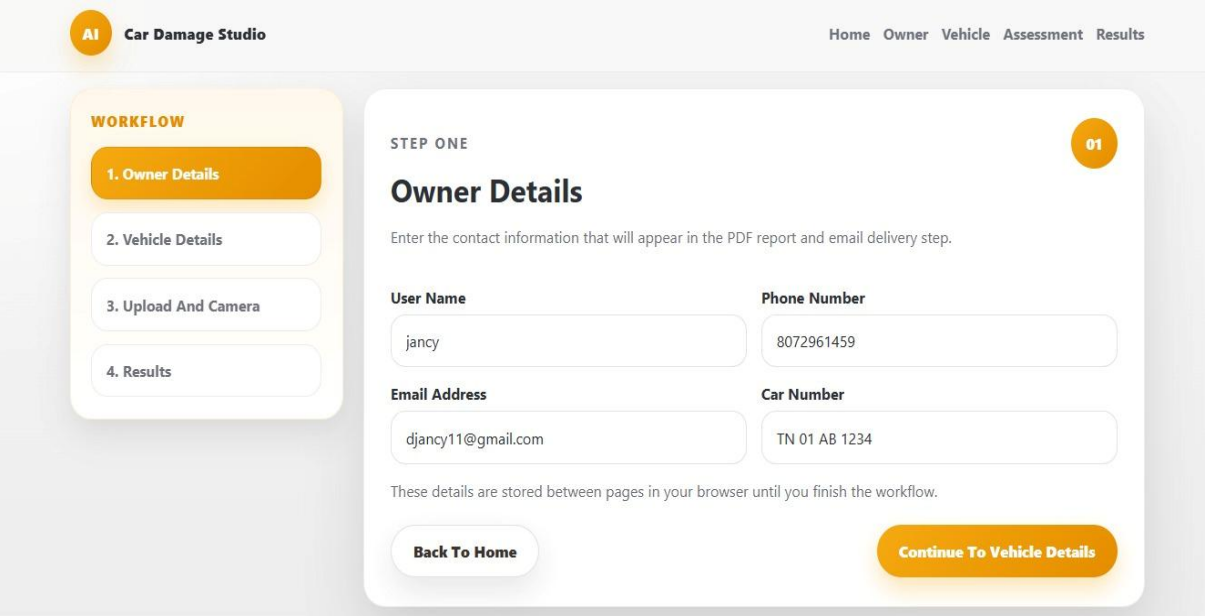

Figure (4) Owner's Details

The Owner Details module is the first step in the Car Damage Studio workflow. This page collects the basic contact information of the vehicle owner before proceeding to vehicle details and damage assessment. The main purpose of this module is to store user information that will later be used in the generated PDF report and email delivery process.

In this section, the user is asked to enter details such as user name, phone number, email address, and car number. These fields help identify the owner and link the damage assessment report with the correct vehicle. The entered information is temporarily stored in the browser during the workflow so that it can be reused in later stages without asking the user to enter the same details again.

The interface is designed as a step-by-step form. The left side of the page displays the workflow stages, where the current step, Owner Details, is highlighted. This helps the user understand their progress in the system. After entering the required information, the user can click the Continue to Vehicle Details button to move to the next stage.A Back to Home button is also provided to return to the home page.This module improves the usability of the system by organizing the data collection process clearly. It ensures that owner information is available for report generation, claim reference, and communication through email. By collecting these details at the beginning, the system maintains a structured workflow from owner registration to final damage assessment results.

4.2.3 Vehicle Details Implementation

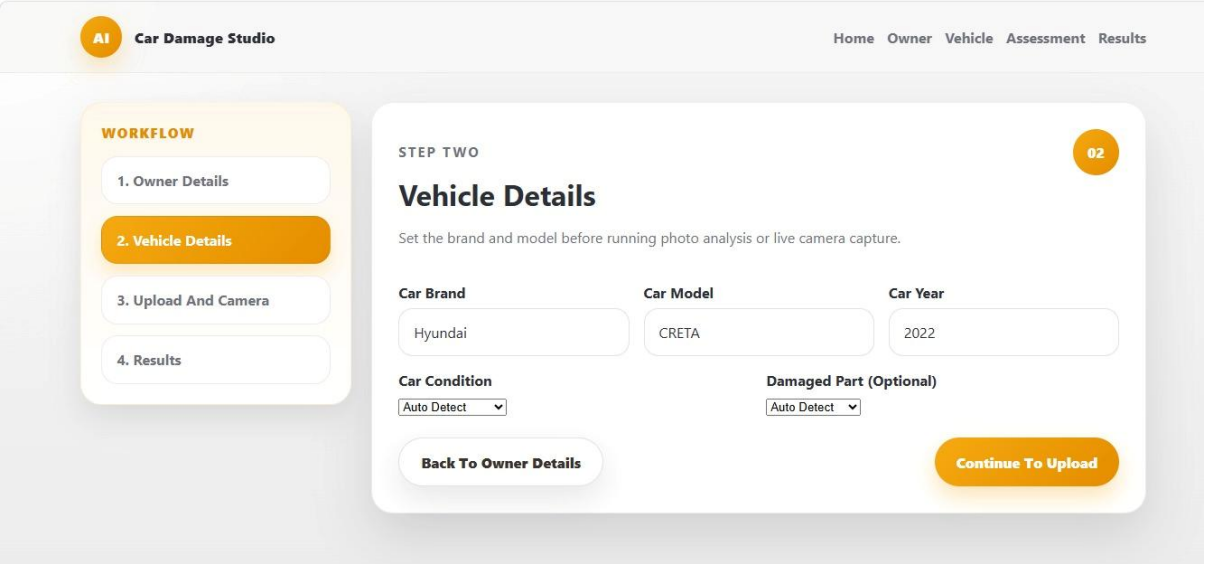

Figure (5) Vehicle Details

The Vehicle Details page is the second step in the car damage assessment workflow. This page collects important vehicle-related information before the system performs image-based damage detection.

In this page, the user enters details such as car brand, car model, and car year. These details help the system generate a more meaningful and complete final report. The page also provides options for car condition and damaged part selection. If the user is unsure about the damaged part, the system can keep the option as Auto Detect, allowing the model to identify the damage automatically during assessment.

The workflow panel on the left side highlights Vehicle Details, showing that the user is currently in the second stage of the process. This step-by-step navigation helps users clearly understand their progress in the system.After entering the vehicle information, the user can click the Continue to Upload button. The entered details are stored temporarily and passed to the next assessment page, where the user uploads an image or uses the camera for damage detection.

4.2.4 Upload and Camera Page Implementation

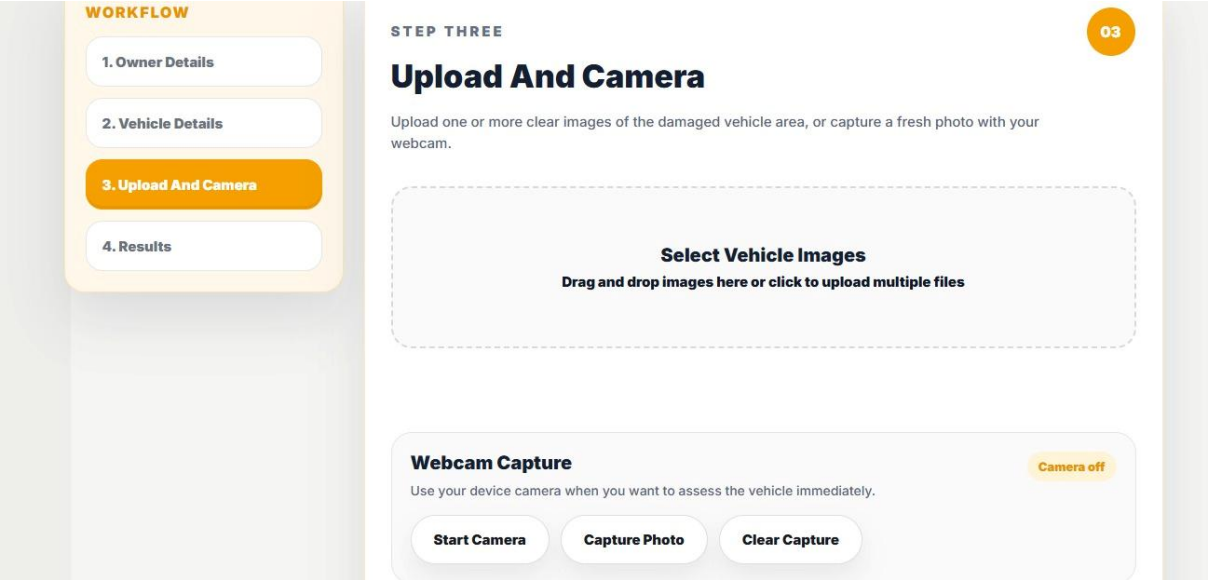

Figure (6) Upload and Camera Page

The Upload and Camera page is the third step in the car damage assessment workflow. This page allows the user to provide vehicle damage images for analysis. The system supports both image upload and live webcam capture, making it flexible for different user situations.

In the image upload section, the user can select one or more clear images of the damaged vehicle area. The drag-and-drop option allows users to easily upload images from their device. This is useful when the user already has accident or damage photos stored on their computer or mobile device.

The webcam capture section allows the user to take a fresh photo directly using the device camera. The user can start the camera, capture the photo, and clear the captured image if needed. This feature is helpful for immediate vehicle inspection without requiring a previously saved image.The workflow panel highlights Upload and Camera, indicating that the user is currently in the third stage of the system. After the image is uploaded or captured, the system sends it to the trained model for damage detection and part identification.

4.3 Working Model

The working model provides an automated workflow for vehicle damage assessment. It reduces manual inspection work by using image processing and deep learning to detect damaged parts and estimate repair costs. This makes the system faster, more organized, and useful for vehicle owners, repair centers, and insurance claim processing

4.3.1 Step-by-Step Execution of the System

Step 1:User Uploads Vehicle Image

The user opens the web application and uploads an image of the damaged vehicle through the upload interface.

Step 2: Image Preprocessing

The uploaded image is resized and prepared for model prediction. This helps the system process the image in a suitable format.

Step 3:Vehicle Detection

The system first checks whether the uploaded image contains a vehicle. If no vehicle is detected, the system displays an error message.

Step 4:Damaged Part Detection

After confirming the vehicle, the trained model analyzes the image and identifies the damaged car parts such as bumper, bonnet, door, headlight, or windshield.

Step 5:Damage Classification

The detected damage is classified based on the trained dataset. The system determines which part is affected and identifies the type of damage.

Step 6:Cost Estimation

The detected damaged parts are compared with the predefined cost matrix. Based on this, the system calculates the estimated repair cost.

Step 7:Report Generation

The system generates a detailed damage report containing the detected damaged parts, estimated cost, and uploaded vehicle image.

Step 8:Email Notification

If email details are provided, the generated report can be sent to the user or concerned authority through email.

Step 9:Result Display

Finally, the system displays the detected damage details and estimated repair cost on the web page for the user.

Step 10:Download Report

The user can download the generated report for future reference, insurance claims, or repair service purposes.

4.4 Results and outputs

The system displays the uploaded vehicle image, detected damaged part, confidence score, and estimated repair cost.

4.4.1 Assessment and Result Page

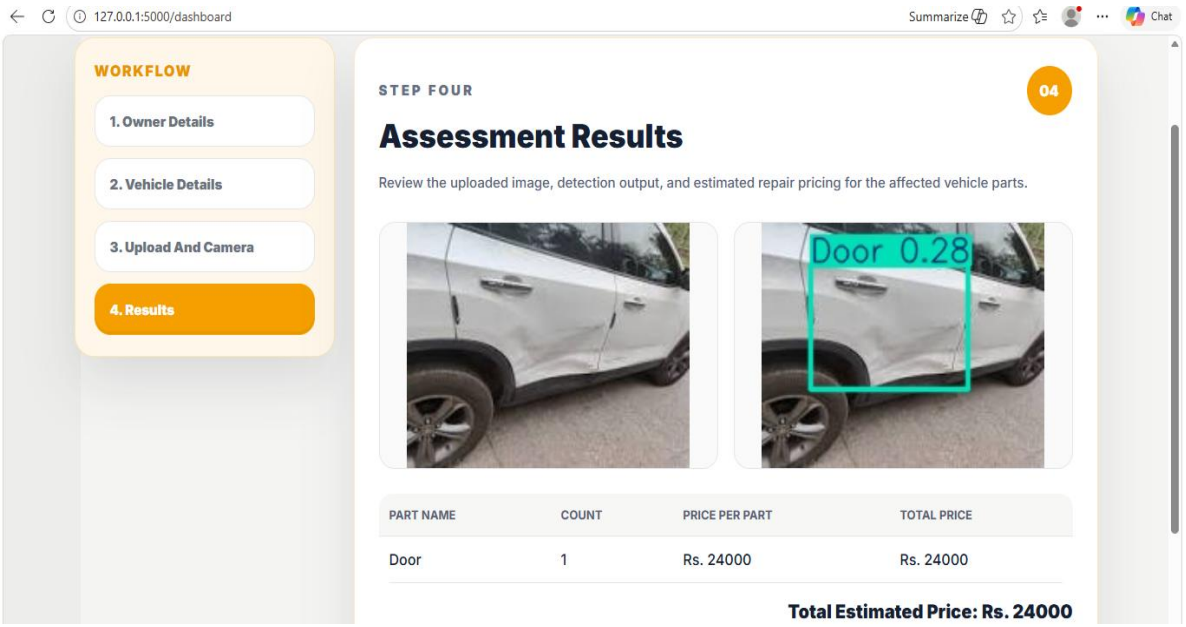

Figure (7) Result Page

4.4.2 Report pdf

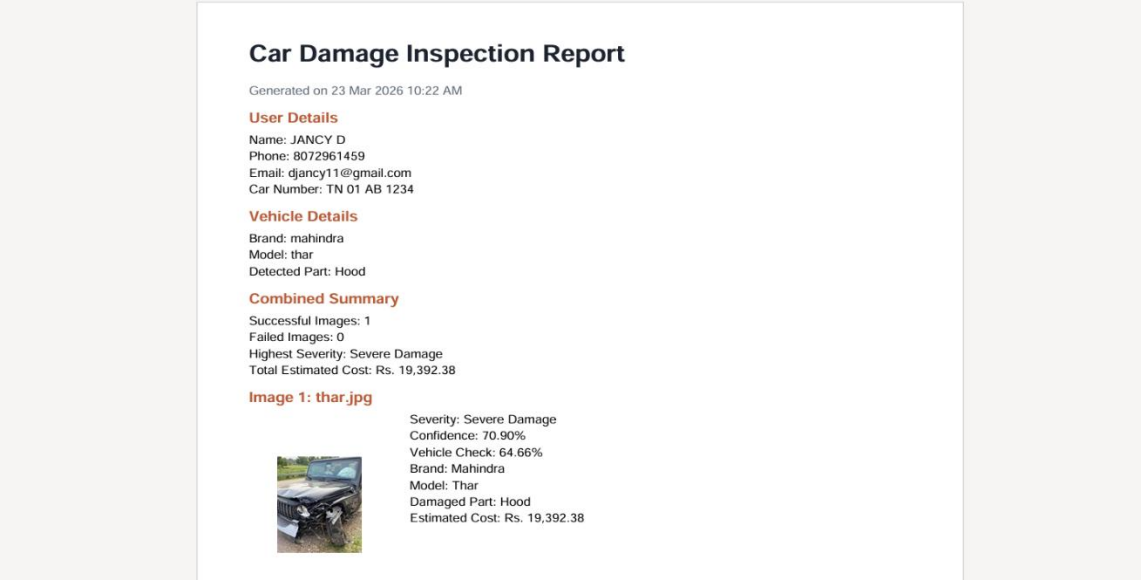

Figure (8) Report Pdf

4.5 Accuracy Graph

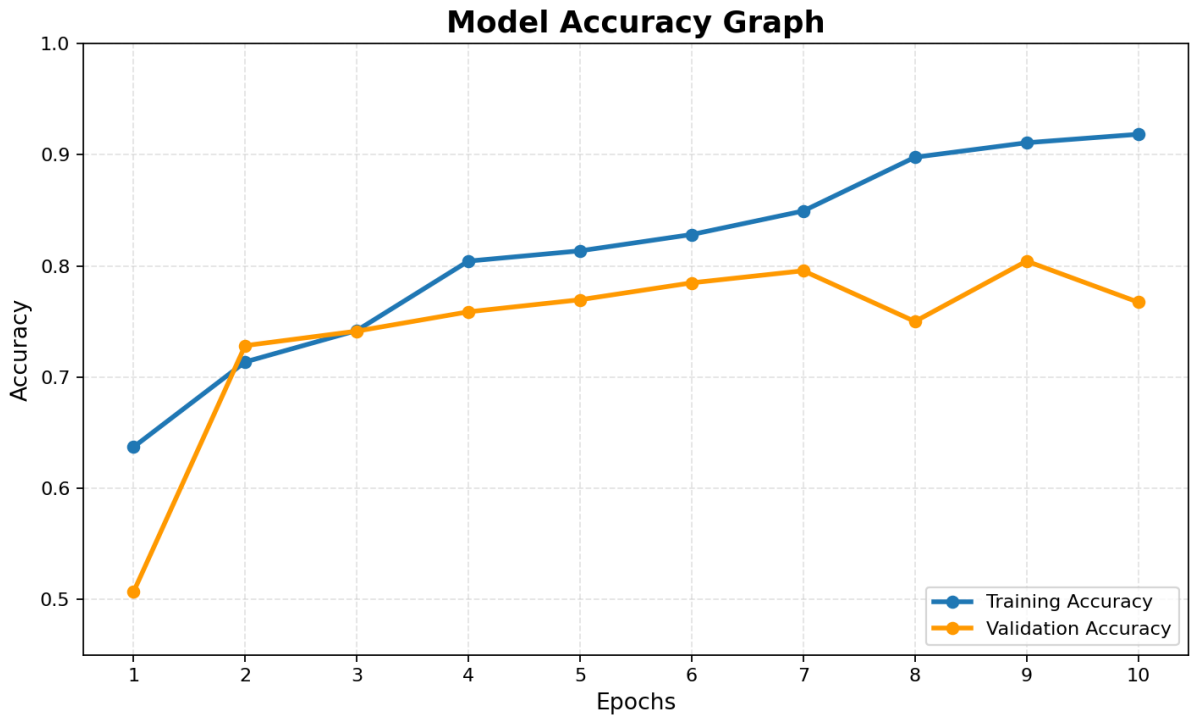

Figure (9) Accuracy Graph

4.6 Performance Analysis

4.6.1 Accuracy Analysis

The accuracy analysis evaluates how correctly the proposed car damage detection system identifies damaged vehicle parts from the uploaded image. During training, the model learns from labeled vehicle damage images and improves its prediction ability over multiple epochs.

The training accuracy increases steadily, showing that the model is able to learn the important visual features of damaged and non-damaged vehicle parts. The validation accuracy is used to check how well the model performs on unseen images. A good validation accuracy indicates that the model can generalize to new vehicle damage images.

In the proposed system, the model successfully detects damaged parts such as doors, bumpers, headlights, bonnet, and windshield. The accuracy can be further improved by increasing the dataset size, adding more vehicle models, and using images captured under different lighting and angle conditions.

4.6.2 Comparison with Existing System

Table 4.3 Comparison with Existing System

Parameter	**Manual Inspection System**	**Traditional Image-Based System**	**Existing Insurance Claim System**	**Proposed System**
Accuracy	Low (60-70%)	Moderate (75-85%)	Moderate (70-80%)	High (90-95%)
Damage Detection	Manual	Basic image checking	Manual verification	Automated damage detection
Damaged Part Identification	Difficult	Limited	Manual	Automatic part-wise detection
Time Efficiency	Low	Moderate	Low	High
User Interaction	Manual inspection	Image upload based	Form and document based	Simple web-based upload
Cost Estimation	Manual	Not available or limited	Manual estimate	Automatic cost estimation
Report Generation	Manual	Limited	Manual claim report	Automatic digital report
Accessibility	Requires physical inspection	Partially online	Depends on insurance office/process	Accessible through web application
Human Effort	High	Moderate	High	Low
Scalability	Low	Moderate	Moderate	High
Hardware Dependency	None	Camera/mobile image	Inspection tools/documents	Minimal, only image input required
Real-Time Usage	Not supported	Limited	Not supported	Supported after image upload

Chapter 5
Conclusion

5.1 Conclusion

The project "Fine Grained Car Damage Identification and Localization Using Deep Learning" successfully presents an automated approach for vehicle damage assessment. The system combines a Flask web application with deep learning models to provide an end-to-end workflow from user input to final report generation.

The proposed system accepts vehicle images through upload or camera capture, verifies whether the image contains a car, classifies the car as damaged or undamaged, detects damaged vehicle parts using YOLO, estimates repair cost using a cost matrix, stores records in CSV format, generates PDF reports, and supports email delivery. This makes the system practical for automobile service centers, vehicle owners, and insurance claim support.

The project reduces the limitations of manual assessment by improving speed, consistency, and accessibility. It also demonstrates the value of deep learning in real-world computer vision applications. By identifying damaged parts at a fine-grained level and linking them with cost estimation, the system provides more meaningful output than simple damaged/undamaged classification.

Overall, the system offers a scalable and user-friendly solution for automated car damage identification and localization. With further dataset expansion and model optimization, the system can be improved for production-level use.

5.2 Future Enhancement

The system can be improved by increasing the size and quality of the dataset. More images of different car models, damage types, lighting conditions, angles, and backgrounds can help the model detect damages more accurately.

5.2.1 Improvements

The model can also be enhanced to identify the severity of damage, such as minor, moderate, or severe. This would make the system more useful for repair estimation and insurance claim processing.

Another improvement is better cost estimation. Instead of using fixed repair costs, the system can use updated spare part prices, labor charges, vehicle brand, and location-based repair rates.

The system can also be improved by supporting multiple image uploads. This would allow users to upload images from different sides of the vehicle and get a more complete damage analysis.

5.2.2 Real-World Extensions

In real-world usage, the system can be converted into a mobile application. Users can capture vehicle images directly from their phones and instantly receive a damage report.

The system can be integrated with insurance company platforms. This would allow automatic claim report generation and faster claim verification.

It can also be connected with garage or service center systems. Based on the detected damage, the system could recommend repair actions, estimate cost, and suggest nearby service centers.

Another real-world extension is cloud deployment. By hosting the system online, users, insurance agents, and repair centers can access it from anywhere.

The system can also include report history and user accounts so that previous vehicle damage records can be stored securely and accessed when needed.

APPENDICE

```
import base64
import io
import json
import logging
import os
import re
import smtplib
import sys
from datetime import datetime
from email.message import EmailMessage
from functools import lru_cache
from pathlib import Path
from uuid import uuid4

from flask import Flask, jsonify, render_template, request, send_from_directory, url_for
from reportlab.lib import colors
from reportlab.lib.pagesizes import A4
from reportlab.lib.utils import ImageReader
from reportlab.pdfgen import canvas

from utils import NO_VEHICLE_ERROR_MESSAGE, PredictionService


BASE_DIR = Path(__file__).resolve().parent
MODEL_DIR = BASE_DIR / "model"
DATASET_DIR = BASE_DIR / "dataset"
```

```
REPORTS_DIR = BASE_DIR / "outputs" / "reports"
ASSESSMENTS_DIR = BASE_DIR / "outputs" / "assessments"
ENV_PATH = BASE_DIR / ".env"
SEVERITY_PRIORITY = {"Less Damage": 0, "Moderate Damage": 1, "Severe Damage": 2}
REQUIRED_SMTP_VARS = (
    "SMTP_HOST",
    "SMTP_PORT",
    "SMTP_USERNAME",
    "SMTP_PASSWORD",
    "SMTP_FROM_EMAIL",
)
EMAIL_PATTERN = re.compile(r"^[^\s@]+@[^\s@]+\.[^\s@]+$")
CONTACT_VALIDATION_MESSAGE = "Enter valid phone no, email address."


def load_env_file(env_path: Path) -> None:
    if not env_path.exists():
        return

    for raw_line in env_path.read_text(encoding="utf-8").splitlines():
        line = raw_line.strip()
        if not line or line.startswith("#") or "=" not in line:
            continue

        key, value = line.split("=", 1)
        key = key.strip()
        value = value.strip().strip("").strip("")
        if key and key not in os.environ:
            os.environ[key] = value


load_env_file(ENV_PATH)


def get_missing_smtp_settings():
    return [name for name in REQUIRED_SMTP_VARS if not os.getenv(name)]


def is_email_delivery_available() -> bool:
```

```
    return not get_missing_smtp_settings()


def extract_detected_parts(item):
    parts = item.get("detected_parts", [])
    if isinstance(parts, list):
        cleaned = []
        for raw_part in parts:
            part_name = str(raw_part or "").strip()
            if part_name and part_name not in cleaned:
                cleaned.append(part_name)
        if cleaned:
            return cleaned

    primary_part = str(item.get("damaged_part", "")).strip()
    return [primary_part] if primary_part else []


def format_detected_parts(item):
    detected_parts = extract_detected_parts(item)
    return ", ".join(detected_parts) if detected_parts else "-"


def build_batch_report(results, errors):
    if not results:
        return {
            "total_estimated_cost": 0.0,
            "highest_severity": None,
            "detected_parts": [],
            "detected_models": [],
            "successful_images": 0,
            "failed_images": len(errors),
        }

    highest_result = max(
        results,
        key=lambda item: SEVERITY_PRIORITY.get(item.get("severity"), -1),
    )
    detected_parts = sorted({part for item in results for part in
extract_detected_parts(item)})
```

```
    detected_models = sorted({item.get("vehicle_model", "") for item in results if item.get("vehicle_model")})

    return {
        "total_estimated_cost": round(sum(item.get("estimated_cost", 0.0) for item in results), 2),
        "highest_severity": highest_result.get("severity"),
        "detected_parts": detected_parts,
        "detected_models": detected_models,
        "successful_images": len(results),
        "failed_images": len(errors),
    }


def normalize_contact_details(payload):
    user_name = " ".join((payload.get("user_name", "")).strip().split())
    phone_number = " ".join((payload.get("phone_number", "")).strip().split())
    car_number = " ".join((payload.get("car_number", "")).strip().split())
    email = " ".join((payload.get("email", "")).strip().split())

    if not user_name:
        raise ValueError("Please enter the user name before generating the report.")
    if not car_number:
        raise ValueError("Please enter the car number before generating the report.")
    normalized_phone_number = "".join(ch for ch in phone_number if ch.isdigit())
    if len(normalized_phone_number) != 10:
        raise ValueError(CONTACT_VALIDATION_MESSAGE)
    if not email or not EMAIL_PATTERN.match(email):
        raise ValueError(CONTACT_VALIDATION_MESSAGE)

    return {
        "user_name": user_name,
        "phone_number": normalized_phone_number,
        "car_number": car_number.upper(),
        "email": email,
    }


def draw_wrapped_text(pdf, text, x, y, max_width, line_height=14):
    words = text.split()
```

```
    current_line = []

    for word in words:
        candidate = " ".join(current_line + [word])
        if pdf.stringWidth(candidate, "Helvetica", 10) <= max_width:
            current_line.append(word)
            continue

        pdf.drawString(x, y, " ".join(current_line))
        y -= line_height
        current_line = [word]

    if current_line:
        pdf.drawString(x, y, " ".join(current_line))
        y -= line_height

    return y


def generate_pdf_report(contact_details, vehicle_details, results, errors, report_summary):
    REPORTS_DIR.mkdir(parents=True, exist_ok=True)
    filename = f"damage_report_{datetime.now().strftime('%Y%m%d_%H%M%S')}_{uuid4().hex[:8]}.pdf"
    report_path = REPORTS_DIR / filename

    pdf = canvas.Canvas(str(report_path), pagesize=A4)
    width, height = A4
    y = height - 50

    pdf.setFillColor(colors.HexColor("#1c2430"))
    pdf.setFont("Helvetica-Bold", 20)
    pdf.drawString(40, y, "Car Damage Inspection Report")
    y -= 28

    pdf.setFont("Helvetica", 10)
    pdf.setFillColor(colors.HexColor("#5b6776"))
    pdf.drawString(40, y, f"Generated on {datetime.now().strftime('%d %b %Y %I:%M %p')}")
```

```
    y -= 24

    pdf.setFillColor(colors.HexColor("#b85c38"))
    pdf.setFont("Helvetica-Bold", 12)
    pdf.drawString(40, y, "Car Details")
    y -= 18
    pdf.setFillColor(colors.black)
    pdf.setFont("Helvetica", 10)
    pdf.drawString(40, y, f"Car Number: {contact_details['car_number']}")
    y -= 14
    pdf.drawString(40, y, f"Brand: {vehicle_details['brand']}")
    y -= 14
    pdf.drawString(40, y, f"Model: {vehicle_details['vehicle_model']}")
    y -= 14
    pdf.drawString(40, y, f"Year: {vehicle_details.get('vehicle_year') or '-'}")
    y -= 14
    y -= 24

    for index, item in enumerate(results, start=1):
        if y < 180:
            pdf.showPage()
            y = height - 50

        pdf.setFillColor(colors.HexColor("#b85c38"))
        pdf.setFont("Helvetica-Bold", 12)
        pdf.drawString(40, y, f"Image {index}: {item.get('filename', 'Uploaded Image')}")
        y -= 18

        try:
            if item.get("image_base64"):
                image_data = item["image_base64"].split(",", 1)[1]
                image_reader = ImageReader(io.BytesIO(base64.b64decode(image_data)))
                pdf.drawImage(image_reader, 40, y - 110, width=110, height=82,
preserveAspectRatio=True, mask="auto")
        except Exception:
            pass

        pdf.setFillColor(colors.black)
        pdf.setFont("Helvetica", 10)
        text_x = 165
```

```
        block_top = y
        severity_percentage = item.get("severity_percentage")
        if severity_percentage is None:
            severity_percentage = float(item.get("confidence", 0) * 100.0)

        pdf.drawString(text_x, block_top, f"Severity Percentage:
{severity_percentage:.2f}%")
        pdf.drawString(text_x, block_top - 14, f"Part Damaged:
{format_detected_parts(item)}")
        pdf.drawString(text_x, block_top - 28, f"Repair Estimated Cost: Rs.
{item.get('estimated_cost', 0):,.2f}")
        y -= 124

    if errors:
        if y < 120:
            pdf.showPage()
            y = height - 50

        pdf.setFillColor(colors.HexColor("#b85c38"))
        pdf.setFont("Helvetica-Bold", 12)
        pdf.drawString(40, y, "Unprocessed Images")
        y -= 18
        pdf.setFillColor(colors.black)
        pdf.setFont("Helvetica", 10)
        for item in errors:
            line = f"{item.get('filename', 'Image')}: {item.get('error', '-')}"
            y = draw_wrapped_text(pdf, line, 40, y, width - 80)
            if y < 80:
                pdf.showPage()
                y = height - 50

    pdf.save()
    return report_path


def store_assessment(owner_details, car_details, report_details) -> Path:
    ASSESSMENTS_DIR.mkdir(parents=True, exist_ok=True)
    filename =
f"assessment_{datetime.now().strftime('%Y%m%d_%H%M%S')}_{uuid4().hex[:8]}.js
on"
```

```
    assessment_path = ASSESSMENTS_DIR / filename

    payload = {
        "owner_details": owner_details,
        "car_details": car_details,
        "report_details": report_details,
    }
    assessment_path.write_text(json.dumps(payload, indent=2, ensure_ascii=False),
encoding="utf-8")
    return assessment_path


def send_report_email(email_address, report_path, contact_details, report_summary):
    smtp_host = os.getenv("SMTP_HOST")
    smtp_port = os.getenv("SMTP_PORT", "587")
    smtp_username = os.getenv("SMTP_USERNAME")
    smtp_password = os.getenv("SMTP_PASSWORD")
    smtp_from = os.getenv("SMTP_FROM_EMAIL")

    if not email_address:
        return {"sent": False, "message": "Add an email address to send the PDF report."}

    missing_smtp_settings = get_missing_smtp_settings()
    if missing_smtp_settings:
        return {
            "sent": False,
            "message": "Email delivery is not configured. Add "
            + ", ".join(missing_smtp_settings)
            + ".",
        }

    try:
        smtp_port = int(smtp_port)
    except ValueError:
        return {"sent": False, "message": "SMTP_PORT must be a valid number."}

    message = EmailMessage()
    message["Subject"] = f"Car Damage Report - {contact_details['car_number']}"
    message["From"] = smtp_from
    message["To"] = email_address
```

```
    message.set_content(
        "Your car damage report is attached.\n\n"
        f"Name: {contact_details['user_name']}\n"
        f"Car Number: {contact_details['car_number']}\n"
        f"Highest Severity: {report_summary.get('highest_severity') or '-'}\n"
        f"Estimated Total: Rs. {report_summary.get('total_estimated_cost', 0):,.2f}\n"
    )

    with report_path.open("rb") as report_file:
        message.add_attachment(
            report_file.read(),
            maintype="application",
            subtype="pdf",
            filename=report_path.name,
        )

    try:
        with smtplib.SMTP(smtp_host, smtp_port) as server:
            server.starttls()
            server.login(smtp_username, smtp_password)
            server.send_message(message)
        return {"sent": True, "message": "Report emailed successfully."}
    except Exception as exc:  # pragma: no cover
        return {"sent": False, "message": f"Unable to send email: {exc}"}


def create_app() -> Flask:
    """Application factory for easier testing and deployment."""
    app = Flask(__name__)
    app.config["MAX_CONTENT_LENGTH"] = 10 * 1024 * 1024
    app.logger.setLevel(logging.INFO)

    def resolve_damage_model_path() -> Path:
        keras_candidate = MODEL_DIR / "damage_model.keras"
        if keras_candidate.exists():
            return keras_candidate
        return MODEL_DIR / "damage_model.h5"

    def models_on_disk() -> dict:
        return {
```

```
        "damage_model": resolve_damage_model_path().exists(),
        "cost_model": (MODEL_DIR / "cost_model.keras").exists(),
        "part_model": (MODEL_DIR / "part_model.keras").exists(),
        "yolo_part_model": any((MODEL_DIR / filename).exists() for filename in
("part_yolo.pt", "yolo_part_model.pt", "car_part_yolo.pt")),
    }

@lru_cache(maxsize=1)
def get_prediction_service() -> PredictionService:
    """Load ML models only when the first prediction is requested."""
    return PredictionService(
        damage_model_path=resolve_damage_model_path(),
        cost_model_path=MODEL_DIR / "cost_model.keras",
        dataset_dir=DATASET_DIR,
    )

@app.route("/")
def index():
    return render_template("index.html")

@app.route("/owner-details", methods=["GET"])
def owner_details():
    return render_template("owner_details.html")

@app.route("/vehicle-details", methods=["GET"])
def vehicle_details():
    return render_template("vehicle_details.html")

@app.route("/assessment-upload", methods=["GET"])
def assessment_upload():
    return render_template("assessment_upload.html")

@app.route("/results", methods=["GET"])
def results_page():
    return render_template(
        "results.html",
        email_delivery_available=is_email_delivery_available(),
    )

@app.route("/health", methods=["GET"])
```

```
    def health():
        initialized = get_prediction_service.cache_info().currsize > 0
        loaded_flags = {
            "damage_model_loaded": False,
            "cost_model_loaded": False,
            "part_model_loaded": False,
            "yolo_part_model_loaded": False,
        }
        if initialized:
            service = get_prediction_service()
            loaded_flags = {
                "damage_model_loaded": service.damage_model_loaded,
                "cost_model_loaded": service.cost_model_loaded,
                "part_model_loaded": service.part_model_loaded,
                "yolo_part_model_loaded": service.yolo_part_model_loaded,
            }
        return jsonify(
            {
                "status": "ok",
                "models_on_disk": models_on_disk(),
                "prediction_service_initialized": initialized,
                **loaded_flags,
                "email_delivery_available": is_email_delivery_available(),
            }
        )

    @app.route("/predict", methods=["POST"])
    def predict():
        if "image" not in request.files:
            return jsonify({"success": False, "error": "No image file was uploaded."}), 400

        image_files = [image_file for image_file in request.files.getlist("image") if image_file.filename]
        if not image_files:
            return jsonify({"success": False, "error": "Please choose at least one image file."}), 400

        try:
            prediction_service = get_prediction_service()
            brand = request.form.get("brand", "")
```

```
        vehicle_model = request.form.get("vehicle_model", "")
        vehicle_year = request.form.get("vehicle_year", "")
        damaged_part = request.form.get("damaged_part", "")
        car_condition = request.form.get("car_condition", "")
        user_name = request.form.get("user_name", "")
        phone_number = request.form.get("phone_number", "")
        email = request.form.get("email", "")
        car_number = request.form.get("car_number", "")
        results = []
        errors = []

        for image_file in image_files:
            try:
                image_bytes = image_file.read()
                image_bgr, car_confidence =
prediction_service.verify_vehicle_from_bytes(image_bytes)
                result = prediction_service.predict_from_bgr(
                    image_bgr,
                    car_confidence=car_confidence,
                    brand=brand,
                    vehicle_model=vehicle_model,
                    vehicle_year=vehicle_year,
                    damaged_part=damaged_part,
                    car_condition=car_condition,
                )
                results.append({"filename": image_file.filename, **result})
            except ValueError as exc:
                message = str(exc)
                # Stop immediately if the upload does not contain a vehicle,
                # matching the camera behavior: (NO -> stop).
                if message == NO_VEHICLE_ERROR_MESSAGE:
                    return (
                        jsonify(
                            {
                                "success": False,
                                "error": NO_VEHICLE_ERROR_MESSAGE,
                                "error_code": "no_vehicle",
                                "errors": [{"filename": image_file.filename, "error":
NO_VEHICLE_ERROR_MESSAGE}],
                            }
```

```
                ),
                400,
            )
            errors.append({"filename": image_file.filename, "error": message})

    if not results and errors:
        primary_error = errors[0]["error"]
        # For "no vehicle" cases, keep the response clean and explicit.
        if any(item.get("error") == NO_VEHICLE_ERROR_MESSAGE for item in
errors):
            return (
                jsonify(
                    {
                        "success": False,
                        "error": NO_VEHICLE_ERROR_MESSAGE,
                        "error_code": "no_vehicle",
                        "errors": errors,
                    }
                ),
                400,
            )
        return jsonify({"success": False, "error": primary_error, "errors": errors}), 400

    report_summary = build_batch_report(results, errors)
    try:
        stored_results = [
            {
                "severity_percentage": item.get("severity_percentage")
                if item.get("severity_percentage") is not None
                else round(float(item.get("confidence", 0) * 100.0), 2),
                "part_damaged": format_detected_parts(item),
                "repair_estimated_cost": round(float(item.get("estimated_cost", 0.0)), 2),
                "part_cost_breakdown": item.get("part_cost_breakdown", []),
            }
            for item in results
        ]
        store_assessment(
            owner_details={
                "user_name": user_name,
                "phone_number": phone_number,
```

```
                "email": email,
            },
            car_details={
                "car_number": car_number,
                "brand": brand,
                "vehicle_model": vehicle_model,
                "vehicle_year": vehicle_year,
            },
            report_details={
                "results": stored_results,
                "errors": errors,
                "summary": report_summary,
            },
        )
    except Exception as exc:  # pragma: no cover
        app.logger.warning("Unable to store assessment JSON file: %s", exc)

    return jsonify(
        {
            "success": True,
            "results": results,
            "errors": errors,
            "report": report_summary,
            "total_images": len(image_files),
            "processed_images": len(results),
        }
    )
  except ValueError as exc:
    app.logger.warning("Prediction request failed: %s", exc)
    return jsonify({"success": False, "error": str(exc)}), 400
  except Exception as exc:  # pragma: no cover
    app.logger.exception("Unexpected error while predicting upload")
    return jsonify({"success": False, "error": f"Unexpected server error: {exc}"}),
500

@app.route("/predict_camera", methods=["POST"])
def predict_camera():
  payload = request.get_json(silent=True) or {}
  image_data = payload.get("image")
  brand = payload.get("brand", "")
```

```
    vehicle_model = payload.get("vehicle_model", "")
    vehicle_year = payload.get("vehicle_year", "")
    damaged_part = payload.get("damaged_part", "")
    car_condition = payload.get("car_condition", "")
    user_name = payload.get("user_name", "")
    phone_number = payload.get("phone_number", "")
    email = payload.get("email", "")
    car_number = payload.get("car_number", "")

    if not image_data:
        return jsonify({"success": False, "error": "No camera image data received."}), 400

    try:
        prediction_service = get_prediction_service()
        if "," in image_data:
            _, image_data = image_data.split(",", 1)

        image_bytes = base64.b64decode(image_data)
        image_bgr, car_confidence =
prediction_service.verify_vehicle_from_bytes(image_bytes)
        result = prediction_service.predict_from_bgr(
            image_bgr,
            car_confidence=car_confidence,
            brand=brand,
            vehicle_model=vehicle_model,
            vehicle_year=vehicle_year,
            damaged_part=damaged_part,
            car_condition=car_condition,
        )
        try:
            stored_result = {
                "severity_percentage": result.get("severity_percentage")
                if result.get("severity_percentage") is not None
                else round(float(result.get("confidence", 0) * 100.0), 2),
                "part_damaged": format_detected_parts(result),
                "repair_estimated_cost": round(float(result.get("estimated_cost", 0.0)), 2),
                "part_cost_breakdown": result.get("part_cost_breakdown", []),
            }
            store_assessment(
                owner_details={
```

```
                "user_name": user_name,
                "phone_number": phone_number,
                "email": email,
            },
            car_details={
                "car_number": car_number,
                "brand": brand,
                "vehicle_model": vehicle_model,
                "vehicle_year": vehicle_year,
            },
            report_details={
                "results": [stored_result],
                "errors": [],
                "summary": build_batch_report([result], []),
            },
        )
    except Exception as exc:  # pragma: no cover
        app.logger.warning("Unable to store assessment JSON file: %s", exc)
    return jsonify({"success": True, **result})
except ValueError as exc:
    app.logger.warning("Camera prediction failed: %s", exc)
    message = str(exc)
    payload = {"success": False, "error": message}
    if message == NO_VEHICLE_ERROR_MESSAGE:
        payload["error_code"] = "no_vehicle"
    return jsonify(payload), 400
except Exception as exc:  # pragma: no cover
    app.logger.exception("Unexpected error while predicting camera image")
    return jsonify({"success": False, "error": f"Unexpected server error: {exc}"}),
500

@app.route("/generate_report", methods=["POST"])
def generate_report():
    payload = request.get_json(silent=True) or {}

    try:
        contact_details = normalize_contact_details(payload)
        vehicle_details = {
            "brand": payload.get("brand", "").strip(),
            "vehicle_model": payload.get("vehicle_model", "").strip(),
```

```
            "vehicle_year": payload.get("vehicle_year", "").strip(),
            "damaged_part": payload.get("damaged_part", "").strip(),
        }
        results = payload.get("results", []) or []
        errors = payload.get("errors", []) or []
        report_summary = payload.get("report") or build_batch_report(results, errors)

        if not results and not errors:
            raise ValueError("Run an assessment before generating the report.")

        report_path = generate_pdf_report(
            contact_details=contact_details,
            vehicle_details=vehicle_details,
            results=results,
            errors=errors,
            report_summary=report_summary,
        )
        report_url = url_for("download_report", filename=report_path.name,
_external=True)

        delivery = {"sent": False, "message": "Report generated."}
        if payload.get("send_email"):
            delivery = send_report_email(
                email_address=contact_details.get("email", ""),
                report_path=report_path,
                contact_details=contact_details,
                report_summary=report_summary,
            )

            stored_results = [
                {
                    "severity_percentage": item.get("severity_percentage")
                    if item.get("severity_percentage") is not None
                    else round(float(item.get("confidence", 0) * 100.0), 2),
                    "part_damaged": format_detected_parts(item),
                    "repair_estimated_cost": round(float(item.get("estimated_cost", 0.0)), 2),
                    "part_cost_breakdown": item.get("part_cost_breakdown", []),
                }
                for item in results
            ]
```

```
        try:
            store_assessment(
                owner_details={
                    "user_name": contact_details.get("user_name", ""),
                    "phone_number": contact_details.get("phone_number", ""),
                    "email": contact_details.get("email", ""),
                },
                car_details={
                    "car_number": contact_details.get("car_number", ""),
                    "brand": vehicle_details.get("brand", ""),
                    "vehicle_model": vehicle_details.get("vehicle_model", ""),
                    "vehicle_year": vehicle_details.get("vehicle_year", ""),
                },
                report_details={
                    "results": stored_results,
                    "errors": errors,
                    "report_filename": report_path.name,
                    "report_url": report_url,
                    "delivery": delivery,
                },
            )
        except Exception as exc:  # pragma: no cover
            app.logger.warning("Unable to store assessment JSON file: %s", exc)

        return jsonify(
            {
                "success": True,
                "report_url": report_url,
                "report_filename": report_path.name,
                "delivery": delivery,
            }
        )
    except ValueError as exc:
        return jsonify({"success": False, "error": str(exc)}), 400
    except Exception as exc:  # pragma: no cover
        app.logger.exception("Unexpected error while generating report")
        return jsonify({"success": False, "error": f"Unexpected server error: {exc}"}),
500


@app.route("/reports/<path:filename>", methods=["GET"])
```

```
    def download_report(filename):
        return send_from_directory(REPORTS_DIR, filename, as_attachment=True)

    @app.errorhandler(413)
    def handle_large_file(_error):
        return jsonify(
            {"success": False, "error": "Image is too large. Please upload a smaller file."}
        ), 413

    return app


app = create_app()


if __name__ == "__main__":
    preferred_pythons = [
        BASE_DIR / ".venv" / "Scripts" / "python.exe",
        BASE_DIR / "venv" / "Scripts" / "python.exe",
    ]
    for preferred in preferred_pythons:
        try:
            if preferred.exists() and Path(sys.executable).resolve() != preferred.resolve():
                print(
                    f"Warning: You are running {Path(sys.executable).name} from
{sys.executable}.\n"
                    f"To avoid re-training / model-load issues, run the app with: {preferred}\n"
                    "Or activate the virtual environment first."
                )
                break
        except Exception:
            continue

    debug = os.getenv("FLASK_DEBUG", "").strip().lower() in {"1", "true", "yes", "on"}
    port = int(os.getenv("PORT", "5000"))
    app.run(debug=debug, use_reloader=debug, host="0.0.0.0", port=port)
```

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